Amal Nambiar

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Professional Summary

Controls & Simulation Engineer with 4+ years validating closed-loop control and trajectory-tracking behavior for ADAS features (AEB, Lane Keeping, ACC) using MATLAB/Simulink and closed-loop virtual simulation. Experienced in sensor fusion and state estimation (Kalman Filter, EKF, Particle Filter), with working knowledge of path-planning and localization algorithms (Dijkstra, A*, RRT, SLAM). Built Python-based tools for scenario-based simulation testing and test automation, with a background in correlating simulated and real-world vehicle trajectory/control behavior.

Experience

Lead Engineer, Mahindra and Mahindra – Chennai, Tamil Nadu **Aug 2022 – Present**

- Validated **closed-loop control logic that tracks planned trajectories** for Lane Keeping Assistant and Adaptive Cruise Control using **MATLAB/Simulink** and **Stateflow**, verifying that control commands accurately followed target paths under varied conditions.
- Built and executed **scenario-based simulation environments** in **SCANeR Studio** to test perception and control algorithms across diverse traffic and environmental conditions, supporting closed-loop algorithm development and validation.
- Applied **Kalman Filter, EKF, and Particle Filter** techniques for radar-camera sensor fusion and state estimation, with working knowledge of **Dijkstra, A*, RRT, and SLAM** algorithms for path planning and localization.
- Conducted **vehicle-level testing on proving track** facilities, correlating real-world trajectory and control behavior against simulation results to validate model accuracy – a back-to-back virtual-to-physical testing approach.
- Developed **Python-based tools** for automated signal correlation (ISO 18571) and test data analysis, reducing manual post-processing effort across simulation-based test campaigns.
- Performed **SIL, HIL, and MIL** test validation across the **V-model** development lifecycle, ensuring feature behavior aligned with **ISO 26262** functional safety and **ASPICE** process compliance.

Research Scholar, IIT Madras – Chennai, Tamil Nadu **Jan 2019 - June 2022**

- Modeled nonlinear dynamical systems in MATLAB using time-series and phase-plane analysis, applying numerical optimization techniques; presented findings at the European Nonlinear Oscillation Conference (ENOC), Lyon, France, 2022.

Education

Indian Institute of Technology Madras – MS in Mechanical Engineering (CGPA 8.6) **Jun 2022**
Sardar Vallabhbhai Institute of Technology, Vasad – BE in Mechanical Engineering (CGPA 8.6) **Jun 2016**

Certifications

ISO 26262 Functional Safety Overview | Advanced Driver Assistance Systems (ADAS) | Kalman Filtering & Sensor Fusion (KF/EKF, Data Fusion) | Applied Control Systems for Autonomous Cars: Math + PID + MPC

Skills

Path Planning & Localization (Working Knowledge): Dijkstra, A*, RRT, SLAM
Control & Trajectory Tracking: MATLAB/Simulink, Stateflow, PID/MPC Concepts
State Estimation & Sensor Fusion: Kalman Filter (KF), Extended Kalman Filter (EKF), Particle Filter, Radar-Camera Fusion
Simulation & Scenario Testing: SCANeR Studio, Closed-loop Virtual Simulation, Scenario-based Testing
Programming: Python, C++ (general), MATLAB
Testing & Process: SIL, HIL, MIL, V-Model / MBD Lifecycle, ISO 26262 Functional Safety, ASPICE